Find a Gene Assignment

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> **Q1.** Tell me the name of a protein you are interested in. Include the species, accession number and known function. This can be a human protein or a protein from any other species as long as it's function is known.
> If you do not have a favorite protein, select human RBP4 or KIF11. Do not use beta globin as this is in the worked example report that I provide you with online.

Name: Interleukin-36 gamma (IL36G)

Species: Homo sapiens

Accession: Q9NZH8

Function: Interleukin-36 gamma (IL36G) plays a role in skin inflammation and issume signaling. IL36G is associated in skin disorders such as psoriasis and dandruff.

> **Q2.** Perform a BLAST search against a DNA database, such as a database consisting of genomic DNA or ESTs. The BLAST server can be at NCBI or elsewhere. Include details of the BLAST method used, database searched and any limits applied (e.g. Organism).

Method: TBLASTN search against reptilia nt

Database: Nucleotide collection (nt)

Organsim: Reptilia (taxid:8459)

> Also include the output of that BLAST search in your document. If appropriate, change the font to Courierdd size 10 so that the results are displayed neatly. You can also screen capture a BLAST output (e.g. alt print screen on a PC or on a MAC press -shift-4. The pointer becomes a bulls eye. Select the area you wish to capture and release. The image is saved as a file called Screen Shot [].png in your Desktop directory). It is not necessary to print out all of the blast results if there are many pages.

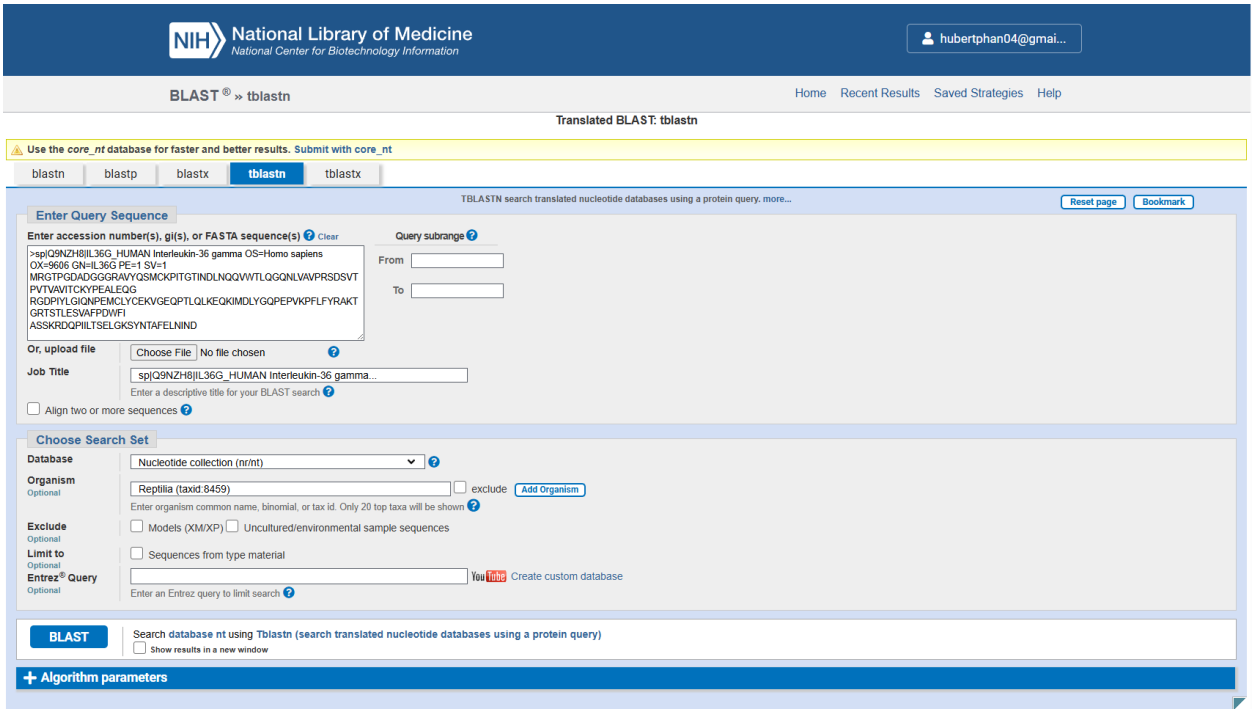

Figure 1: TBLASTN QUERY

On the BLAST results, clearly indicate a match that represents a protein sequence, encoded from some DNA sequence, that is homologous to your query protein. I need to be able to inspect the pairwise alignment you have selected, including the E value and score. It should be labeled a "genomic clone" or "mRNA sequence", etc. - but include no functional annotation.

Chosen Match: Accession OZ346534.1, Trionyx triunguis genome assembly. See alignment details below.

Edit Search | Save Search | Search Summary | How to read this report? | BLAST Help Videos | Back to Traditional Results Page

Your search is limited to records **include: Reptilia (taxid:8459)**

Job Title	**sp|Q9NZH8|IL36G_HUMAN Interleukin-36 gamma...**
RID	RU2HMSBS016 *Search expires on 02-01 02:44 am* Download All
Program	TBLASTN Citation
Database	nt See details
Query ID	lcl|Query_6676515
Description	sp|Q9NZH8|IL36G_HUMAN Interleukin-36 gamma OS=Hor ...
Molecule type	amino acid
Query Length	169
Other reports	

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to **E value** to **Query Coverage** to

Filter **Reset**

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments **Download** **Select columns** Show 100

select all *73 sequences selected* GenBank Graphics

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
PREDICTED: Chelonoidis abingdonii interleukin-36 receptor antagonist protein-like (LOC116819404), m...	Chelonoidis abi...	95.5	95.5	90%	1e-20	35.29%	3011	XM_032771100.2
PREDICTED: Emys orbicularis interleukin-36 receptor antagonist protein-like (LOC135873734), mRNA	Emys orbicularis	87.8	87.8	89%	1e-19	35.53%	576	XM_065398370.1
PREDICTED: Chelonoidis abingdonii interleukin-36 receptor antagonist protein-like (LOC116819405), m...	Chelonoidis abi...	87.8	87.8	89%	1e-19	34.87%	576	XM_032771101.1
PREDICTED: Gopherus flavomarginatus interleukin-36 receptor antagonist protein-like (LOC127044660)...	Gopherus flavo...	87.4	87.4	89%	2e-19	34.87%	576	XM_050939738.1
PREDICTED: Malaclemys terrapin pileata interleukin-36 receptor antagonist protein-like (LOC12883275...	Malaclemys terr...	90.1	90.1	85%	2e-19	36.81%	1090	XM_054020339.1
PREDICTED: Mauremys reevesii interleukin-36 receptor antagonist protein-like (LOC120398387), transc...	Mauremys reev...	88.2	88.2	84%	3e-19	37.06%	764	XM_039525790.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123364511), transcri...	Mauremys mutica	90.9	90.9	90%	3e-19	35.95%	2253	XM_045006715.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123364511), transcri...	Mauremys mutica	90.9	90.9	90%	3e-19	35.95%	2267	XM_045006714.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123364511), transcri...	Mauremys mutica	90.9	90.9	90%	3e-19	35.95%	2207	XM_045006713.1
PREDICTED: Chrysemys picta bellii interleukin-36 receptor antagonist protein (LOC101949597), mRNA	Chrysemys pict...	90.1	180	85%	4e-19	36.81%	1357	XM_005279282.4
PREDICTED: Trachemys scripta elegans interleukin-36 receptor antagonist protein-like (LOC117871587...	Trachemys scri...	90.1	90.1	85%	5e-19	36.81%	1389	XM_034759399.1
PREDICTED: Mauremys reevesii interleukin-36 receptor antagonist protein-like (LOC120398387), transc...	Mauremys reev...	88.2	88.2	84%	5e-19	37.06%	836	XM_039525788.1
PREDICTED: Emydura macquarii macquarii interleukin-36 receptor antagonist protein-like (LOC1371479...	Emydura macq...	87.8	87.8	96%	6e-19	32.75%	821	XM_067540356.1
PREDICTED: Mauremys reevesii interleukin-36 receptor antagonist protein-like (LOC120398387), transc...	Mauremys reev...	88.2	88.2	84%	6e-19	37.06%	896	XM_039525787.1
PREDICTED: Carettochelys insculpta interleukin-36 receptor antagonist protein-like (LOC142004348), m...	Carettochelys in...	85.5	85.5	91%	7e-19	34.19%	529	XM_074981942.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123365076), transcri...	Mauremys mutica	88.2	88.2	82%	1e-18	36.69%	1072	XM_045007705.1
PREDICTED: Emydura macquarii macquarii interleukin-36 receptor antagonist protein-like (LOC1371478...	Emydura macq...	83.6	83.6	90%	2e-17	33.33%	787	XM_067540265.1
PREDICTED: Trachemys scripta elegans interleukin-36 receptor antagonist protein-like (LOC117871587...	Trachemys scri...	82.4	82.4	85%	2e-16	34.84%	1434	XM_034759398.1
PREDICTED: Pelodiscus sinensis interleukin-1 family member 10-like (LOC142820928), transcript varia...	Pelodiscus sine...	79.0	79.0	82%	4e-15	35.25%	1692	XM_075910708.1
PREDICTED: Pelodiscus sinensis interleukin-1 family member 10-like (LOC142820928), transcript varia...	Pelodiscus sine...	79.0	79.0	79%	5e-15	35.82%	1690	XM_075910709.1
PREDICTED: Pelodiscus sinensis interleukin-1 family member 10-like (LOC142820928), transcript varia...	Pelodiscus sine...	78.6	78.6	79%	6e-15	35.82%	1857	XM_075910706.1
PREDICTED: Mauremys mutica interleukin-36 receptor antagonist protein-like (LOC123365076), transcri...	Mauremys mutica	75.5	75.5	53%	4e-14	43.96%	949	XM_045007706.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC102448715), trans...	Pelodiscus sine...	75.1	75.1	90%	9e-14	31.37%	1468	XM_075910776.1
PREDICTED: Carettochelys insculpta interleukin-36 receptor antagonist protein-like (LOC142004315), m...	Carettochelys in...	72.4	72.4	85%	2e-13	32.88%	653	XM_074981886.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC102448715), trans...	Pelodiscus sine...	73.6	73.6	90%	3e-13	31.37%	1428	XM_075910775.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	71.6	71.6	88%	1e-12	31.13%	1290	XM_075910855.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	71.2	71.2	88%	2e-12	31.13%	1290	XM_075910852.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	71.2	71.2	88%	2e-12	31.13%	1180	XM_075910854.1
PREDICTED: Pelodiscus sinensis interleukin-36 receptor antagonist protein-like (LOC142820938), trans...	Pelodiscus sine...	70.9	70.9	88%	2e-12	31.13%	1283	XM_075910851.1

Figure 2: TBLASTN RESULTS

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Trionyx triunguis genome assembly, chromosome: 30

Sequence ID: OZ346534.1 Length: 17913073 Number of Matches: 1

Range 1: 1420032 to 1420238 GenBank Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
50.1 bits(118)	5e-05	Compositional matrix adjust.	26/69(38%)	41/69(59%)	1/69(1%)	+3

```
Query  91       VGEQPTLQLKEQKIMDLYGQPEPVKPFLFYRAKTGRTSTLESVAFPDWFIASSKRD-QPI  149
                +   P    ++++IMDLY   +  K F F    TG T   ESVAFP WF+++S +D QP+
Sbjct  1420032  LSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGWFLSTSPKDGQPL  1420211

Query  150      ILTSELGKS  158
                 +T+  G++
Sbjct  1420212  RVTNISGEA  1420238
```

Figure 3: Alignment Details

In general, [Q2] is the most difficult for students because it requires you to have a "feel" for how to interpret BLAST results. You need to distinguish between a perfect match to your query (i.e. a sequence that is not "novel"), a near match (something that might be "novel", depending on the results of [Q4]), and a non-homologous result.
If you are having trouble finding a novel gene try restricting your search to an organism that is poorly annotated.

Q3. Gather information about this "novel" protein. At a minimum, show me the protein sequence of the "novel" protein as displayed in your BLAST results from [Q2] as FASTA format (you can copy and paste the aligned sequence subject lines from your BLAST result page if necessary) or translate your novel DNA sequence using a tool called EMBOSS Transeq at the EBI. Don't forget to translate all six reading frames; the ORF (open reading frame) is likely to be the longest sequence without a stop codon. It may not start with a methionine if you don't have the complete coding region. Make sure the sequence you provide includes a header/subject line and is in traditional FASTA format.

The genomic DNA region surrounding the tblastn hit in Trionyx triunguis was extracted from the genome assembly and expanded beyond the aligned coordinates to capture potential coding sequence. The sequence was then translated in all six reading frames using EMBOSS Transeq. The longest open reading frame (ORF) without a stop codon was identified by examining for conserved motifs observed in the original BLAST alignment. This segment is the protein sequence used for downstream BLASTP analysis to assess sequence novelty.

Chosen Sequence:

```
>Trionyx_triunguis_tblastn_hit_translated_fragment frame_2
GQWADGRSLPPSLQAVSPFLIGLSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGW
FLSTSPKDGQPLRVTNISGEAEIINFYFKKM
```

Here, tell me the name of the novel protein, and the species from which it derives. It is very unlikely (but still definitely possible) that you will find a novel gene from an organism such as S. cerevisiae, human or mouse, because those genomes have already

been thoroughly annotated. It is more likely that you will discover a new gene in a genome that is currently being sequenced, such as bacteria or plants or protozoa.

Name: IL-1 family-like protein

Species: Trionyx triunguis (African softshell turtle)

Lineage: Eukaryota; Metazoa; Chordata; Reptilia; Testudines; Trionychidae; Trionyx

Q4. For the purposes of this project, "novel" is defined as follows. Take the protein sequence (your answer to [Q3]) and use it as a query in a BLASTP search of the nr database at NCBI.

- If there is a match with **100% amino acid identity** to a protein in the database **from the same species**, then your protein is **NOT novel** (even if the match is to a protein with a name such as "unknown"). Someone has already found and annotated this sequence and assigned it an accession number.
- If the **top match reported has less than 100% identity**, then it is likely that your protein is **novel**, and you have succeeded.
- If there is a match with **100% identity**, but to a **different species** than the one you started with, then you have likely succeeded in finding a novel gene.
- If there are **no database matches** to the original query from [Q1], this indicates that you have **partially succeeded**: yes, you may have found a new gene, but no, it is not actually homologous to the original query. You should probably start over.

Details:

A BLASTP search against the nr database yielded a top hit protein from *Pelodiscus sinensis* (Chinese softshell turtle).

Below are screenshots of the BLASTP query, BLASTP results, and alignment details of the top hit.

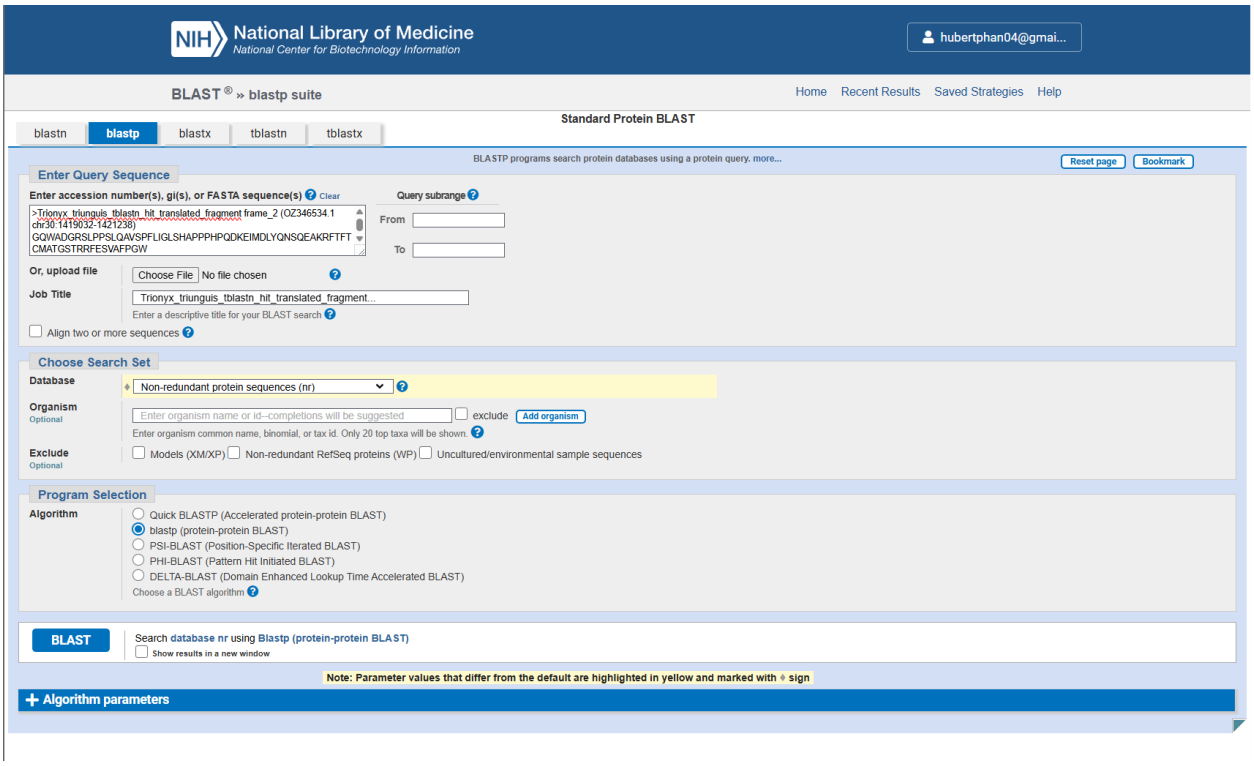

Figure 4: BLASTP Query

Edit Search | Save Search | Search Summary | How to read this report? | BLAST Help Videos | Back to Traditional Results Page

Job Title	Trionyx_triunguis_tblastn_hit_translated_fragment...
RID	RV67KW6H014 *Search expires on 02-01 12:53 pm* Download All
Program	BLASTP Citation
Database	nr See details
Query ID	lcl|Query_10407620
Description	Trionyx_triunguis_tblastn_hit_translated_fragment frame_2 ...
Molecule type	amino acid
Query Length	101
Other reports	Distance tree of results Multiple alignment MSA viewer

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to **E value** to **Query Coverage** to

Filter Reset

Compare these results against the new Clustered nr database BLAST

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download Select columns Show 100

select all *100 sequences selected* GenPept Graphics Distance tree of results Multiple alignment MSA Viewer

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
interleukin-1 family member 10-like isoform X4 [Pelodiscus sinensis]	Pelodiscus sinensis	141	141	79%	3e-39	85.00%	236	XP_075766824.1
interleukin-1 family member 10-like isoform X3 [Pelodiscus sinensis]	Pelodiscus sinensis	142	142	80%	4e-39	83.95%	255	XP_075766823.1
uncharacterized protein LOC142820928 isoform X2 [Pelodiscus sinensis]	Pelodiscus sinensis	142	142	80%	8e-39	83.95%	302	XP_075766822.1
uncharacterized protein LOC142820928 isoform X1 [Pelodiscus sinensis]	Pelodiscus sinensis	142	142	69%	2e-38	92.86%	344	XP_075766821.1
interleukin-36 receptor antagonist protein-like [Chelonoidis abingdonii]	Chelonoidis abingdonii	124	124	80%	6e-33	74.07%	191	XP_032626992.1
interleukin-36 receptor antagonist protein-like [Emys orbicularis]	Emys orbicularis	124	124	79%	6e-33	75.00%	191	XP_065254442.1
interleukin-36 receptor antagonist protein-like isoform X2 [Trachemys scripta elegans]	Trachemys scripta elegans	125	125	80%	1e-32	75.31%	275	XP_034615290.1
interleukin-36 receptor antagonist protein-like [Malaclemys terrapin pileata]	Malaclemys terrapin pileata	125	125	80%	1e-32	75.31%	275	XP_053876314.1
interleukin-36 receptor antagonist protein [Chrysemys picta bellii]	Chrysemys picta bellii	124	124	80%	2e-32	74.07%	254	XP_005279339.2
interleukin-1 receptor antagonist protein-like isoform X1 [Trachemys scripta elegans]	Trachemys scripta elegans	125	125	80%	3e-32	75.31%	290	XP_034615289.1
tetracycline resistance MFS efflux pump [Platysternon megacephalum]	Platysternon megacephalum	122	122	79%	1e-31	73.75%	236	TFJ97238.1
interleukin-36 receptor antagonist protein-like [Gopherus flavomarginatus]	Gopherus flavomarginatus	120	120	80%	2e-31	72.84%	191	XP_050795695.1
hypothetical protein KIL84_011473 [Mauremys mutica]	Mauremys mutica	121	121	79%	2e-31	72.50%	219	KAH1177771.1
interleukin-36 receptor antagonist protein-like isoform X2 [Mauremys mutica]	Mauremys mutica	120	120	80%	2e-31	71.60%	200	XP_044863641.1
interleukin-36 receptor antagonist protein-like isoform X1 [Mauremys mutica]	Mauremys mutica	120	120	80%	4e-31	71.60%	241	XP_044863640.1
unnamed protein product [Caretta caretta]	Caretta caretta	112	112	69%	1e-28	74.29%	150	CAM4704126.1

Figure 5: BLASTP Results

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interleukin-1 family member 10-like isoform X4 [Pelodiscus sinensis]

Sequence ID: XP_075766824.1 Length: 236 Number of Matches: 1

Range 1: 157 to 236 GenPept Graphics ▼Next Match ▲Previous Match

Score	Expect	Method	Identities	Positives	Gaps
141 bits(356)	3e-39	Compositional matrix adjust.	68/80(85%)	72/80(90%)	0/80(0%)

```
Query  22   GLSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGWFLSTSPKDGQP  81
            G S  P    +DKEIM LYQNSQEAKRFTFTC+ATGSTRRFESVAFPGWFLSTSPKDGQP
Sbjct  157  GTSAPPTLMLEDKEIMALYQNSQEAKRFTFTCVATGSTRRFESVAFPGWFLSTSPKDGQP  216

Query  82   LRVTNISGEAEIINFYFKKM  101
            +RVTNISGEAEI+NFYFKKM
Sbjct  217  VRVTNISGEAEIVNFYFKKM  236
```

Related Information
Genome Data Viewer - aligned genomic context

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interleukin-1 family member 10-like isoform X3 [Pelodiscus sinensis]

Sequence ID: XP_075766823.1 Length: 255 Number of Matches: 1

Range 1: 175 to 255 GenPept Graphics ▼Next Match ▲Previous Match

Score	Expect	Method	Identities	Positives	Gaps
142 bits(357)	4e-39	Compositional matrix adjust.	68/81(84%)	72/81(88%)	0/81(0%)

```
Query  21   IGLSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGWFLSTSPKDGQ  80
             G S  P    +DKEIM LYQNSQEAKRFTFTC+ATGSTRRFESVAFPGWFLSTSPKDGQ
Sbjct  175  CGTSAPPTLMLEDKEIMALYQNSQEAKRFTFTCVATGSTRRFESVAFPGWFLSTSPKDGQ  234

Query  81   PLRVTNISGEAEIINFYFKKM  101
            P+RVTNISGEAEI+NFYFKKM
Sbjct  235  PVRVTNISGEAEIVNFYFKKM  255
```

Related Information
Genome Data Viewer - aligned genomic context

Figure 6: Alignment Details

[Q5] Generate a multiple sequence alignment with your novel protein, your original query protein, and a group of other members of this family from different species. A typical number of proteins to use in a multiple sequence alignment for this assignment purpose is a minimum of 5 and a maximum of 20 - although the exact number is up to you. Include the multiple sequence alignment in your report. Use Courier font with a size appropriate to fit page width. Side-note: Indicate your sequence in the alignment by choosing an appropriate name for each sequence in the input unaligned sequence file (i.e. edit the sequence file so that the species, or short common, names (rather than accession numbers) display in the output alignment and in the subsequent answers below). The goal in this step is to create an interesting an alignment for building a phylogenetic tree that illustrates species divergence.

Re-labeled sequences for alignment:

```
>Trionyx_triunguis IL-1 family-like protein frame_2 (novel protein)
GQWADGRSLPPSLQAVSPFLIGLSHAPPPHPQDKEIMDLYQNSQEAKRFTFTCMATGSTRRFESVAFPGW
FLSTSPKDGQPLRVTNISGEAEIINFYFKKM

>IL36G_HUMAN
MRGTPGDADGGGRAVYQSMCKPITGTINDLNQQVWTLQGQNLVAVPRSDSVTPVTVAVIT
CKYPEALEQGRGDPIYLGIQNPEMCLYCEKVGEQPTLQLKEQKIMDLYGQPEPVKPFLFY
RAKTGRTSTLESVAFPDWFIASSKRDQPIILTSELGKSYNTAFELNIND

>Pelodiscus_sinensis XP_075766824.1 interleukin-1 family member 10-like isoform X4
MVSRTDNGIRATSPLREGAAEEDSFPSQAPETSTDVGMGSSRPAPPQDTAGTEPWPGAGPRPDSWAQPSN
RDLVELFDEFLNQKQQFIPSVPKHSLFTLRDTNQRVIRRLHKQLVATPQTSNAPPEKISVVPNQFLDPSN
FPIIMGIDNGTRCLSCGTSAPPTLMLEDKEIMALYQNSQEAKRFTFTCVATGSTRRFESVAFPGWFLSTS
PKDGQPVRVTNISGEAEIVNFYFKKM
```

```
>Chelonoidis_abingdonii XP_032626992.1 interleukin-36 receptor antagonist protein-like
MTSTGHTSPAWPGAGRTPGPYSQPDNSNLVDLFNKFFSQRPAIMTVPGPSLFTLRDTSQKVVRCHHNRLM
ASPQTANTLPEKISVVPNQFMDPSHFPIIMGIDGGTRCLSCGTSAQPTLMLEDKKIMDLYQNSQEAKRFT
FTCSATGSTHRFESVAFPGWYLSTSPRNDQPLQVTNRLGEAEITNFYFKKV

>Emys_orbicularis XP_065254442.1 interleukin-36 receptor antagonist protein-like
MTGTGHTTPARPGAGRTPGSYSQPGNSNLVNLFNEFFKQRPTIMAVPGPSLFTLRDTSQKVVRYHHNRLV
ASPQTANAPPEKISVVPNQFMDPSHFPIIMGINGGTRCLSCGTSAQPTLMLEDKKIMDLYQNSREAKRFT
FTCSATGSTLRFESVAFPGWYLSTSPRNDQPLRVTNRLGEAEITNFYFKKV

>Trachemys_scripta_elegans XP_034615290.1 interleukin-36 receptor antagonist protein-like isoform X2
MVSSLLLALLLTCCSFDCFSSQSMTNNGIWATPLPGEDETEEDAFLSQVRKTSTDLGLDSNDPEGPVSAA
GSASTHTPPCSHQDMTGTGNMSPAQPGAGRTPGPYSQPGNSNLVNLFNEFFKQGPTIMAVPRPSLFTLRD
TSQKVVRCHHNHLVASPQTANAPPEKISVVPNQFMDPSHFPIIMGINGGTRCLSCGTSAQPTLMLEDKKI
MDLYQNSQEAKRFTFTCSATGSTLRFESVAFPGWYLSTSPRNDQPLRVTNRLGEAEITNFYFKKV

>Chelonia_mydas XP_043392104.1 interleukin-36 receptor antagonist protein
MSISDIPANAADAGVLAQANVPRDQEELRLSRDSCSVFPEKTSVVPNQFMDPSHFPIVMGIDGGLCCLSC
STSAQPTLMLEDKKIMDLYQNSWEAERFTFTCSASSSTLRFELVAFPGWYLSTSPRNDQPLWVTYHLGEA
EITNFYFKKV

>Podarcis_cretensis XP_079811393.1 interleukin-36 receptor antagonist protein-like isoform 1
MEKVDTETLREKPHKMTVDQKMRDLFHHFPKPDHSKFPPISLDRPWLYRIWDINQKFLFLKNNMLVAAPK
DGNSPDYLVAVTPNRGMDENKSPIFLGTQDGAQTLSCGESGGQPQLTLEHKAIMDLYNDGKEHKNFTFFC
KSGSSTETGSFESAAFPGWFLSTLPEPNQPIRLSHQGGAEITQFYFDKVKD

>Dromiciops_gliroides XP_043845630.1 interleukin-36 receptor antagonist protein
MVLQKALCFRMKDAALKVIYVQNNQLLARRVQAGKPIDGEEISVVPNRSLDAKRSPVILGLEGGTQCLSV
GIAQEPVLQIEHKNIMDLYRSKEESKSFTFYMWDTGLTSRLESAAYPGWFLCTIHEDEKPITLTNHPEDS
ESVITDFYFHQCA

>Myotis_myotis XP_036167553.1 interleukin-36 receptor antagonist protein
MVLSGALCFRMKDAALKVLYLQDNQLLAGGVHAGKAVKGEEISVVPNRFLDDSLSPVILGVQGGSQCLSC
GMGQEPTLKLEPVDIMELYRSPKDSRGFTFYRRDTGLTSRFESAAFPGWFLCTVPEADQPLRLSQLPGDA
SWDHPIMDFYFQQCD
```

MSA:

```
CLUSTAL O(1.2.4) multiple sequence alignment


IL36G_HUMAN                 ------------------------------------------------------------ 0
Podarcis_cretensis          ------------------------------------------------------------ 0
Trionyx_triunguis           ------------------------------------------------------------ 0
Pelodiscus_sinensis         --------------------MVSRTDNGIRATSPLREGAAEEDSFPSQAPETSTDVGMGS 40
Chelonia_mydas              ------------------------------------------------------------ 0
Chelonoidis_abingdonii      ------------------------------------------------------------ 0
Emys_orbicularis            ------------------------------------------------------------ 0
Trachemys_scripta_elegans   MVSSLLLALLLTCCSFDCFSSQSMTNNGIWATPLPGEDETEEDAFLSQVRKTSTDLGLDS 60
Dromiciops_gliroides        ------------------------------------------------------------ 0
Myotis_myotis               ------------------------------------------------------------ 0


IL36G_HUMAN                 -------------------------------MRGTPGDA----------DGGGRAV---- 15
Podarcis_cretensis          -------------------------------MEKVDTETLREKPHKMTVDQKMRDLFHHF 29
Trionyx_triunguis           ------------------------------------------------------------ 0
Pelodiscus_sinensis         SRPAPPQDTAGT--------------------EPWPGAGPRPDSWAQPSNRDLVELFDEF 80
```

```
Chelonia_mydas                 ------------------------------------------------------------ 0
Chelonoidis_abingdonii         ------------------------MTSTGHTSPAWPGAGRTPGPYSQPDNSNLVDLFNKF 36
Emys_orbicularis               ------------------------MTGTGHTTPARPGAGRTPGSYSQPGNSNLVNLFNEF 36
Trachemys_scripta_elegans      NDPEGPVSAAGSASTHTPPCSHQDMTGTGNMSPAQPGAGRTPGPYSQPGNSNLVNLFNEF 120
Dromiciops_gliroides           ------------------------------------------------------------ 0
Myotis_myotis                  ------------------------------------------------------------ 0


IL36G_HUMAN                    ---------YQSMCKPITGTINDLNQQVWTLQGQNLVAVPRSD--SVTPVTVAVITCKYP 64
Podarcis_cretensis             PKPDHSKFPPISLDRPWLYRIWDINQKFLFLKNNMLVAAPKDG--NSPDYLVAVT----P 83
Trionyx_triunguis              ------------------------------------------------------------ 0
Pelodiscus_sinensis            LNQKQQ--FIPSVPKHSLFTLRDTNQRVIRRLHKQLVATPQTS--NAPPEKISVV----P 132
Chelonia_mydas                 ----M---SISDIPANAA-DAGVLAQANVPRDQEELRLSRDSC--SVFPEKTSVV----P 46
Chelonoidis_abingdonii         FSQRP---AIMTVPGPSLFTLRDTSQKVVRCHHNRLMASPQTA--NTLPEKISVV----P 87
Emys_orbicularis               FKQRP---TIMAVPGPSLFTLRDTSQKVVRYHHNRLVASPQTA--NAPPEKISVV----P 87
Trachemys_scripta_elegans      FKQGP---TIMAVPRPSLFTLRDTSQKVVRCHHNHLVASPQTA--NAPPEKISVV----P 171
Dromiciops_gliroides           ----------MVLQKALCFRMKDAALKVIYVQNNQLLARRVQAGKPIDGEEISVV----P 46
Myotis_myotis                  ----------MVLSGALCFRMKDAALKVLYLQDNQLLAGGVHAGKAVKGEEISVV----P 46


IL36G_HUMAN                    EALEQGRGDPIYLGIQNPEMCLYCEKVGEQPTLQLKEQKIMDLYGQPEPVKPFLFYRAKT 124
Podarcis_cretensis             NRGMDENKSPIFLGTQDGAQTLSCGESGGQPQLTLEHKAIMDLYNDGKEHKNFTFFCKSG 143
Trionyx_triunguis              GQWADGRSLPPSLQ---AVSPFLIGLS-HAPPPHPQDKEIMDLYQNSQEAKRFTFTCMAT 56
Pelodiscus_sinensis            NQFLDPSNFPIIMGIDNGTRCLSCGTS-APPTLMLEDKEIMALYQNSQEAKRFTFTCVAT 191
Chelonia_mydas                 NQFMDPSHFPIVMGIDGGLCCLSCSTS-AQPTLMLEDKKIMDLYQNSWEAERFTFTCSAS 105
Chelonoidis_abingdonii         NQFMDPSHFPIIMGIDGGTRCLSCGTS-AQPTLMLEDKKIMDLYQNSQEAKRFTFTCSAT 146
Emys_orbicularis               NQFMDPSHFPIIMGINGGTRCLSCGTS-AQPTLMLEDKKIMDLYQNSREAKRFTFTCSAT 146
Trachemys_scripta_elegans      NQFMDPSHFPIIMGINGGTRCLSCGTS-AQPTLMLEDKKIMDLYQNSQEAKRFTFTCSAT 230
Dromiciops_gliroides           NRSLDAKRSPVILGLEGGTQCLSVGIA-QEPVLQIEHKNIMDLYRSKEESKSFTFYMWDT 105
Myotis_myotis                  NRFLDDSLSPVILGVQGGSQCLSCGMG-QEPTLKLEPVDIMELYRSPKDSRGFTFYRRDT 105
                                   :    *  :        :        *    :   ** ** .    . * *

IL36G_HUMAN                    GR--TSTLESVAFPDWFIASSK-RDQPIILTSELGKSY----NTAFELNIND-    169
Podarcis_cretensis             SSTETGSFESAAFPGWFLSTLPEPNQPIRLSHQGGA-----EITQFYFDKVKD    191
Trionyx_triunguis              GS--TRRFESVAFPGWFLSTSPKDGQPLRVTNISGEA----EIINFYFKKM--    101
Pelodiscus_sinensis            GS--TRRFESVAFPGWFLSTSPKDGQPVRVTNISGEA----EIVNFYFKKM--    236
Chelonia_mydas                 SS--TLRFELVAFPGWYLSTSPRNDQPLWVTYHLGEA----EITNFYFKKV--    150
Chelonoidis_abingdonii         GS--THRFESVAFPGWYLSTSPRNDQPLQVTNRLGEA----EITNFYFKKV--    191
Emys_orbicularis               GS--TLRFESVAFPGWYLSTSPRNDQPLRVTNRLGEA----EITNFYFKKV--    191
Trachemys_scripta_elegans      GS--TLRFESVAFPGWYLSTSPRNDQPLRVTNRLGEA----EITNFYFKKV--    275
Dromiciops_gliroides           GL--TSRLESAAYPGWFLCTIHEDEKPITLTNHPEDS--ESVITDFYFHQCA-    153
Myotis_myotis                  GL--TSRFESAAFPGWFLCTVPEADQPLRLSQLPGDASWDHPIMDFYFQQCD-    155
                             .   *  :* .*:*.*::.:     :*: ::              * :.
```

The multiple sequence alignment demonstrates that the novel Trionyx triunguis fragment clusters with IL36RN-like proteins across reptiles and mammals. Highly conserved motifs including IMDLYQNSQEAKRFTFTC and FESVAFPGWFLSTSPK are shared with IL36RN orthologs but differ from IL36G, indicating that the novel sequence is most closely related to the IL-36 receptor antagonist subgroup of the IL-1 family.

> [Q6] Create a phylogenetic tree, using either a parsimony or distance-based approach. Bootstrapping and tree rooting are optional. Use "simple phylogeny" online from the EBI or any respected phylogeny program (such as MEGA, PAUP, or Phylip). Paste an image of your Cladogram or tree output in your report.

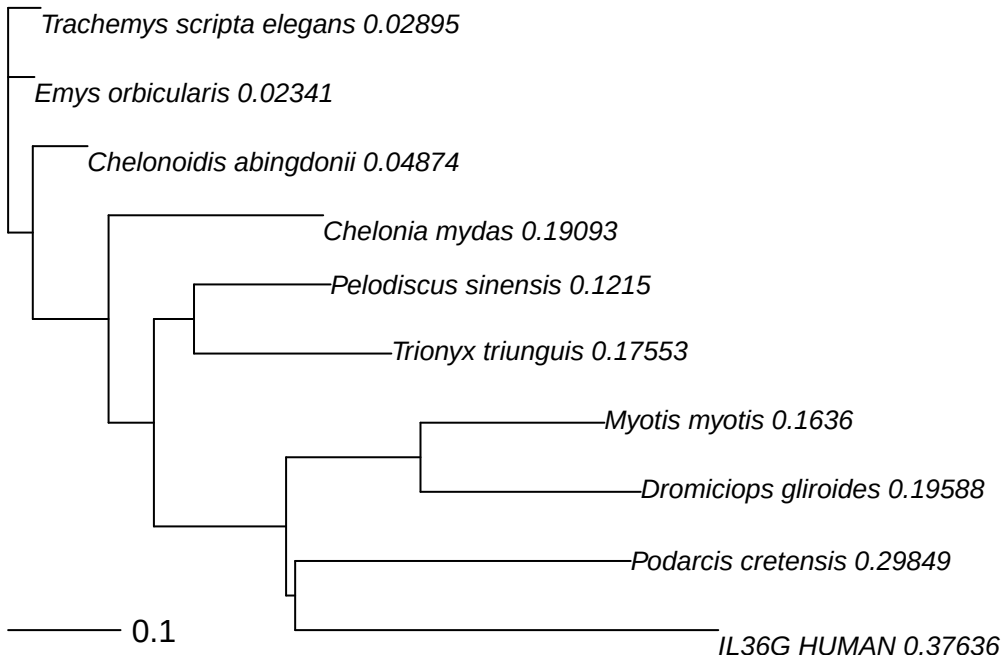

Figure 7: Phylogenetic Tree

[Q7] Generate a sequence identity based heatmap of your aligned sequences using R. If necessary convert your sequence alignment to the ubiquitous FASTA format (Seaview can read in clustal format and "Save as" FASTA format for example). Read this FASTA format alignment into R with the help of functions in the Bio3D package. Calculate a sequence identity matrix (again using a function within the Bio3D package). Then generate a heatmap plot and add to your report. Do make sure your labels are visible and not cut at the figure margins.

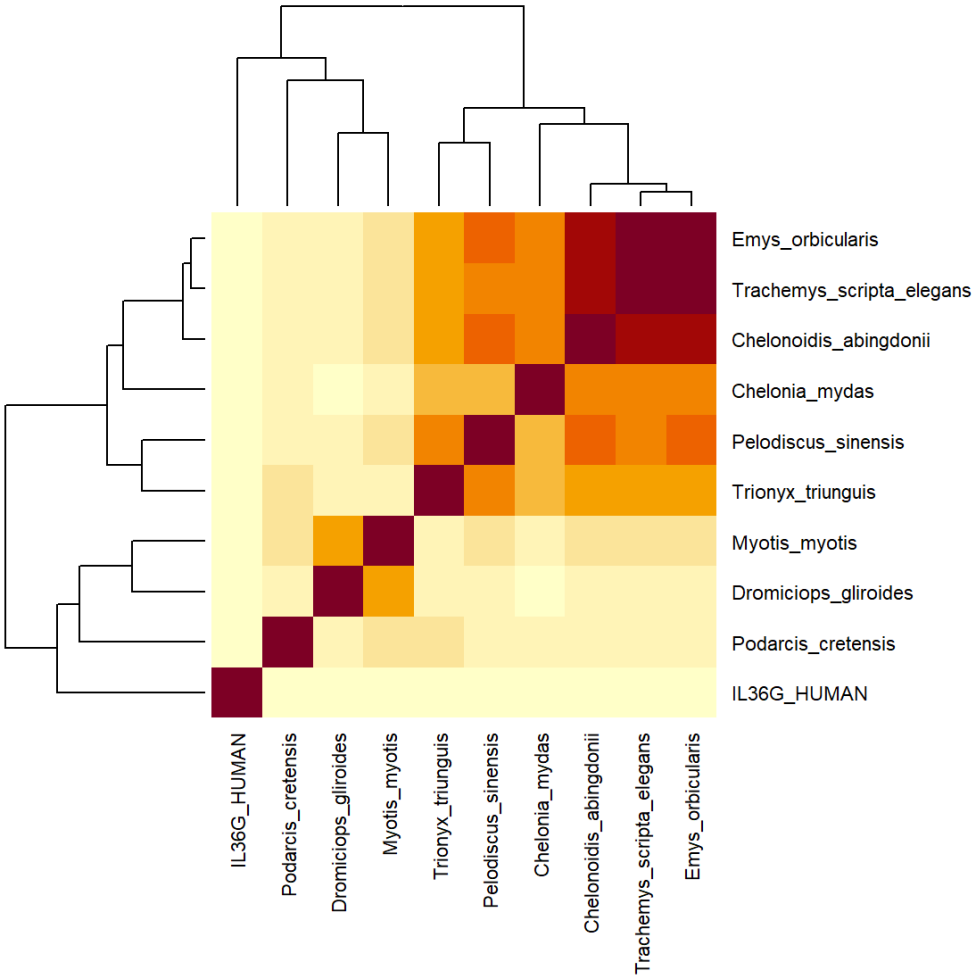

Figure 8: Clustered Heatmap

[Q8] Using R/Bio3D (or an online blast server if you prefer), search the main protein structure database for the most similar atomic resolution structures to your aligned sequences.

List the top 3 unique hits (i.e. not hits representing different chains from the same structure) along with their Evalue and sequence identity to your query. Please also add annotation details of these structures. For example include the annotation terms PDB identifier (structureId), Method used to solve the structure (experimentalTechnique), resolution (resolution), and source organism (source).

HINT: You can use a single sequence from your alignment or generate a consensus sequence from your alignment using the Bio3D function consensus(). The Bio3D functions blast.pdb(), plot.blast() and pdb.annotate() are likely to be of most relevance for completing this task. Note that the results of blast.pdb() contain the hits PDB identifier (or pdb.id) as well as Evalue and identity. The results of pdb.annotate() contain the other annotation terms noted above. Note that if your consensus sequence has lots

of gap positions then it will be better to use an original sequence from the alignment for your search of the PDB. In this case you could chose the sequence with the highest identity to all others in your alignment by calculating the row-wise maximum from your sequence identity matrix.

```
    ID Technique Resolution       Source   Evalue Identity
1 4P0J     X-RAY       2.30 Homo sapiens 2.12e-42     44.6
2 1MD6     X-RAY       1.60 Mus musculus 1.02e-38     44.1
3 4P0L     X-RAY       1.55 Homo sapiens 2.62e-38     41.8
```